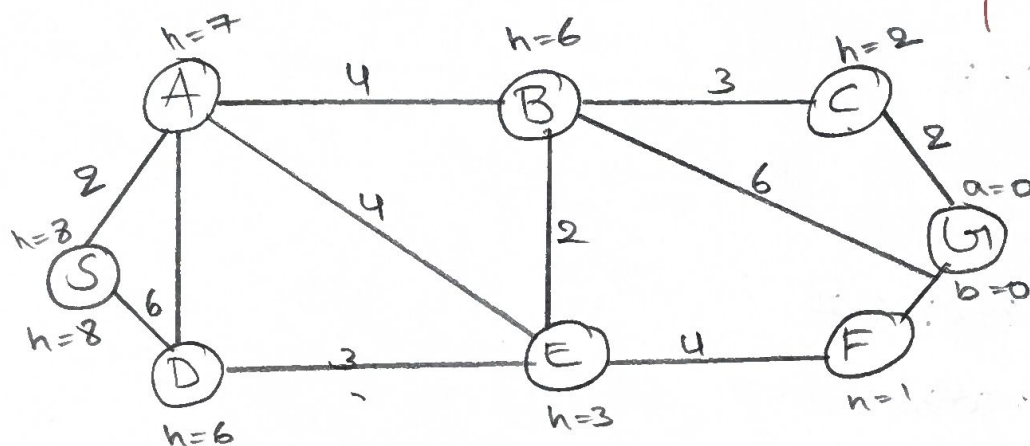


CO2: Assessment Tool-2Q1. Smart Ambulance Navigation:

Given,

Starting Point = S

Goal = G

A* algorithm uses:

$$f(n) = g(n) + h(n)$$

Where

 $g(n)$ = cost from S to current node $h(n)$ = estimated costStep 1:Start from S $\rightarrow$ A or D

For A

$$g(A) = 2, h(A) = 7$$

$$f(A) = 2 + 7 = 9$$

For D

$$g(D) = 6, h(D) = 5$$

$$f(D) = 6 + 5 = 11$$

 $\therefore$ Select A

98

49

10/8/26

Step 2:

From A $\rightarrow$ B or D or E

For B:

$$g(B) = 2 + 4 = 6$$

$$f(B) = 6 + 6 = 12$$

For E:

$$g(E) = 2 + 4 = 6$$

$$f(E) = 6 + 3 = 9$$

$\therefore$ Select E

Step 3:

From E $\rightarrow$ B, D & F

~~The~~ The best path

B $\rightarrow$ C $\rightarrow$ G

For C:

$$g(C) = 6 + 3 = 9$$

$$h(C) = 2$$

$$f(C) = 9 + 2 = 11$$

From C to G

$$h(G) = 0$$

$$f(G) = 11$$

$\therefore$ Goal Reached

Final Route:

S $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ G

Total cost:

$$2 + 4 + 3 + 2 = 11$$

Autonomous Warehouse Robot:

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

S
G
#
?

Start

Goal

Obstacle

Unknown cell

Given,

The robot moves from the charging station (S) to the Target Shelf (G)

The important feature is that the robot does not initially know all obstacles. It discovers obstacles when it reaches nearby cells

Let's use a online search Method

Heuristic used

$$h(n) = |x - x_g| + |y - y_g|$$

Step by step:

The robot starts from S.

One possible path is

$S \rightarrow (2,5) \rightarrow (3,5)$

Then it checks the next cells and moves

$(3,5) \rightarrow (3,4) \rightarrow (3,3)$

Then

$(3,3) \rightarrow (3,2) \rightarrow (4,2)$

Then $(4,2) \rightarrow (5,2) \rightarrow (5,1)$

$\therefore$ The final path is

$S \rightarrow (2,5) \rightarrow (3,5) \rightarrow (3,4) \rightarrow (3,3) \rightarrow (3,2) \rightarrow (4,2) \rightarrow (5,2) \rightarrow (5,1) \rightarrow G$

* Online Search vs Offline Search:

Online Search:

- The complete environment is not known.
- Obstacles are discovered while moving.
- The path can be changed when a new obstacle are found.
- It is useful for real-world robots.

Offline Search:

- The complete environment is already known.
- The path is planned before movement.
- It is suitable when map is already available.

* Online search is suitable for the warehouse robot because it can detect new obstacles and change its path while moving.

University Examination Scheduling:

Examination Hall	Capacity
Hall A	120 St
Hall B	90 St
Hall C	60 St

(a) CSP Formulation:

CSP $\rightarrow$ Constraint Satisfaction Problem

$\rightarrow$ It contains:-

(i) Variables: $E_1, E_2, E_3, E_4, E_5, E_6$

(ii) Domains: T_1, T_2, T_3, T_4

(iii) Constraints:

$$E_1 \neq E_2$$

$$E_1 \neq E_3$$

$$E_2 \neq E_4$$

$$E_3 \neq E_5$$

$$E_4 \neq E_6$$

$$E_2 \neq E_5$$

Therefore

E_1 can use Hall A

E_2 can use Hall A or Hall B

E_3 can use Hall A or B or C

E_4 can use Hall A or B or C

E_5 can use Hall A or B

E_6 can use Hall A or B or C

(b) MRV, Backtracking and Forward ch.

All exams time slot:

T_1, T_2, T_3, T_4

Step 1:

Select E_2 and Assign

$$E_2 = T_1$$

$$E_1 = [T_2, T_3, T_4]$$

$$E_2 = [T_2, T_3, T_4]$$

$$E_5 = [T_2, T_3, T_4]$$

Step 2:

$$E_5 = T_2$$

Step 3:

$$E_3 = T_3$$

Step 4:

$$E_4 = T_3$$

Step 5:

$$E_6 = T_1$$

Final Time Table:

$$T_1 \rightarrow E_2, E_6$$

$$T_2 \rightarrow E_5, E_1$$

$$T_3 \rightarrow E_3, E_4$$

$$T_4 \rightarrow \text{No exam.}$$

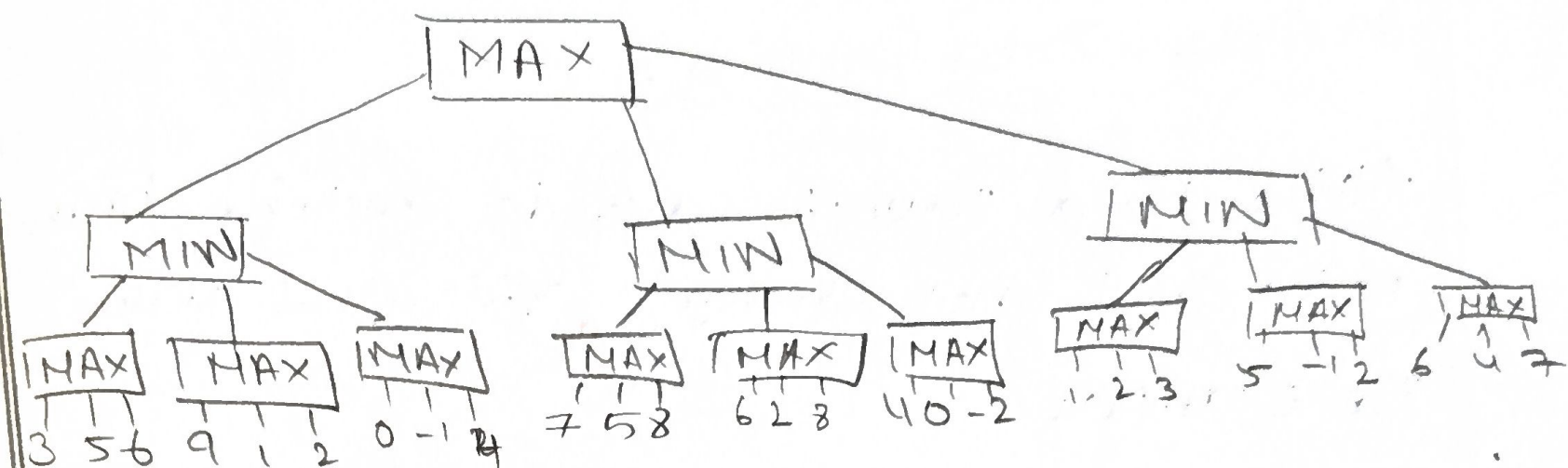
Hall Allocation:

$$T_1: E_2 \rightarrow \text{Hall A}, E_6 \rightarrow \text{Hall B}$$

$$T_2: E_5 \rightarrow \text{Hall B}, E_1 \rightarrow \text{Hall A}$$

$$T_3: E_3 \rightarrow \text{Hall B}, E_4 \rightarrow \text{Hall C}$$

Game playing AI:



Minimax is used to find the best move in a game. MAX selects the highest values and MIN selects the lowest value

Step 1:

$$\text{Max}(3, 5, 6) = 6$$

$$\text{Max}(9, 1, 2) = 9$$

$$\text{Max}(0, -1, 4) = 4$$

$$\text{Min}(6, 9, 4) = 4$$

Step 2:

$$\text{Max}(7, 5, 8) = 8$$

$$\text{Max}(6, 2, 3) = 6$$

$$\text{Max}(4, 0, 2) = 4$$

$$\text{Min}(8, 6, 4) = 4$$

Step 3:

$$\text{Max}(1, 2, 3) = 3$$

$$\text{Max}(5, 1, 2) = 5$$

$$\text{Max}(6, 4, 7) = 7$$

$$\text{Min}(3, 5, 7) = 3$$

Step 4:

$$\text{Max}(4, 4, 3) = 4$$

* Alpha - Beta Pruning:

Alpha-beta pruning avoids checking branches that will not change the final value

For example:

In the left branch after getting value 6
Some remaining values can be skipped

50 Delivery Route Optimization:

Step 1

$$S_1 = 28 \text{ km}$$

$$S_5 = 25 \text{ km}$$

$$\text{Since } 25 < 28$$

Step 2 Move to 85

Let

$$22 < 25$$

Move in this order.

Step 3

$$\text{Final distance} = 22 \text{ km}$$

The problem is a travelling salesman problem where the objective is to minimize the total distance travelled while visiting all customers and returning to the starting point D

The initial solution is

$$S_1 = 26 \text{ km}$$

For minimization, hill climbing chooses the neighboring solution with the lowest cost

Among

$$26, 30, 24, 27, 22, 29$$

The minimum is

$$22 \text{ km}$$

$\therefore$ New Solution:

~~D $\rightarrow$ A $\rightarrow$ E $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D~~

Cost:

$$22 \text{ km}$$

* And it is Not Global Optimum

$\therefore$ Hill Climbing improves the initial route from 26 km to 22 km and selects.

~~D $\rightarrow$ A $\rightarrow$ E $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D~~

With a cost of

$$22 \text{ km}$$